

ESP32 CYD MASTER FLASHING GUIDE

Deployment & Diagnostic Reference for Cheap Yellow Display Boards (2.8" & 4.0")

LAB REF #CYD-01

01 HARDWARE BRIDGE & DRIVER MATRIX

Connection timeouts and missing COM/TTY ports are overwhelmingly caused by charge-only USB cables or missing USB-to-UART bridge drivers. Verify your board's serial chip before attempting to flash firmware.

BRIDGE CHIP	TARGET BOARD MODEL	REQUIRED DRIVER	OS DEVICE MANAGER IDENTIFIER
`CH340 / CH341`	Standard 2.8" Micro-USB / Type-C	WCH CH341SER Driver (v3.8+)	`USB-SERIAL CH340 (COMX)`
`CP2102 / CP2104`	Premium 2.8" & Dual USB-C CYDs	Silicon Labs CP210x VCP (v6.7+)	`Silicon Labs CP210x...`
`CH9102F`	4.0" ST7796 & High-Speed CYDs	WCH CH343/CH9102 VCP Driver	`USB-SERIAL CH9102 (COMX)`

Critical Cable Notice: Always verify your USB cable using a smartphone or external drive to ensure it contains internal data lines. Over 40% of initial setup support tickets stem from using 2-wire power-only charging cords.

02 MANUAL BOOT MODE SEQUENCES

Due to variations in auto-reset circuitry across manufacturer revisions, automatic flashing tools frequently fail. Select the hardware boot sequence tching oard behavior:

A. Standard Toggle Sequence

(Best for standard 2.8" single-USB with auto-reset lines)

1. Connect CYD to PC via USB data cable.
2. Press & hold BOOT button.
3. Press & release `RESET (EN)` button once.
4. Release BOOT button. Click **Install** on flasher.

B. Pre-Power Boot Sequence

(Best for boards missing GPIO0 pull-down resistors)

1. Unplug USB cable from the board.
2. Press & hold BOOT button while unplugged.
3. Plug in USB cable while maintaining hold.
4. Release BOOT button after 2 seconds.

C. Sustained Hold Sequence

(High-Failure Rate Boards)

Symptom: Flashing drops mid-write at 5%–30% with *"Timed out waiting for packet header"*.

Procedure: Press and hold the BOOT button before initiating the upload in your web flasher or Arduino IDE, and maintain the hold continuously throughout the entire write cycle until completion.

03 WEB FLASHER & IDE FLASHING CONFIGURATION

When flashing pre-compiled binaries (e.g., ESP32 Marauder, SquatchWatch, LVGL GUIs) or compiling from source, set parameters strictly to match display memory constraints:

BOARD TARGET
`ESP32 Dev Module`
Standard ESP32-WROOM-32 base pin layout

FLASH FREQUENCY
`80MHz (Lower to 40MHz if erratic)`
Sustains high display refresh without bus errors

04 HARDWARE DIAGNOSTIC & RECOVERY WORKFLOWS

Issue: Permanent Black Screen After Flash

GPIO Pin Mismatch: Verify LCD Backlight pin is explicitly set to `GPIO 21` (Standard 2.8") or `GPIO 27` (4.0" ST7796).
SPI Drivers: Ensure `TFT_eSPI` configuration uses the correct `ST7789` driver for 2.8" panels or `ST7796` for 4.0" display panels.

Issue: Continuous Boot Loop (Brownout)

Power Supply: ESP32 Wi-Fi bursts draw up to 500mA transient current. Avoid unpowered USB hubs; connect directly to motherboard USB ports.
Corrupted NVS: Perform a full flash erase using `esptool.py erase_flash` before rewriting firmware binaries.

Issue: Touchscreen Calibration Fix (XPT2046)

If touch inputs are inverted, unregistered, or misaligned, execute a dedicated XPT2046 calibration sketch to write boundary offsets (`X_MIN`, `X_MAX`, `Y_MIN`, `Y_MAX`) directly to EEPROM. Ensure SPI touch Chip Select (CS) pin is mapped to `GPIO 33`.

05 LAB BENCH FINDINGS & KNOWN HARDWARE QUIRKS

USB-C Power Delivery (PD) Negotiation Failure (0V Rail Output)

Observed Quirk: Connecting budget CYD boards directly to modern USB-C Power Delivery (PD) chargers (e.g., Anker Nano 30W GaN) via a direct USB-C to USB-C cable results in 0V output and an unpowered board.

Root Cause: PCB layout omits mandatory 5.1kΩ pull-down resistors on CC1 and CC2 lines. Smart PD supplies detect floating pins and refuse to initialize the 5V rail.

Lab Workaround: Power the board using a standard USB-A to USB-C cable or an inline USB-C to USB-A adapter.

Display Driver Glass Panel Identification (TPM408-28 vs. H685)

Generic Budget Units: Display glass marked with `TPM408-28` typically utilizes the **ILI9341** driver.

Specialty Units (e.g., NerdMiner): Display glass marked with `H685` typically utilizes the **ST7789** driver.

Lab Warning: Flashing incorrect display driver headers causes a persistent white screen. Always verify your display glass text before compiling or flashing firmware.

06 DEPLOYMENT CHECKLIST FOR ZERO CUSTOMER RETURNS

1. Pre-Flashing Cleanse

Execute full erase flash when transitioning between different firmware frameworks.

2. Driver Verification

Confirm COM port appears in Device Manager prior to launching flash utilities.

3. Button Mechanics

Test Sustained Hold Sequence if standard boot toggle yields packet header timeouts.

4. Asset Integration

Provide full partition maps, driver links, and hardware quirk notices with deployment assets.